

This report analyzes air quality trends in counties from 1999 to 2023. The goal of this report is to find areas of concern, patterns, and improvements over this time period. Some of the key metrics used are Air Quality Index (AQI), unhealthy days, and comparison across counties and states. All the data can be found from the Environmental Protection Agency (EPA).

Some good news is that over the years air quality has improved in America. Figure 2 shows that the average AQI has improved from 1999 to 2023. It also consistently shows that Summer is the season with the worst air quality, followed by spring. Winter and fall typically have the same level of air quality with negligible differences.

Even with the overall improvement in America, some areas have gone backwards in air quality. The counties that saw the largest decline in air quality are found in North Dakota, Idaho, Montana, Minnesota, Colorado, Wyoming and one county in Puerto Rico. Burke in North Dakota was the county with the largest change with an AQI decline of 41.6. See figure 1 for a visual representation.

Are the lowest ranked counties getting better? Figure 3 shows that California is the only state that has constantly had multiple counties that ranked in the top 10 for the worst AQI ranking. Also important to note that there has been little change in the AQI scores in the bottom 10 counties, however the counties for each time period that are the bottom 10 do seem to change.

What states have shown improvement and how much? Figure 4 shows that Kentucky has had the largest improvement of all the states in air quality. Despite all the bad news shown about California in figure 3, it is ranked number 4 in the most improved states, however it still towers above the other top 10 states with their present day average AQI score.

Utah was found not to be in the top 10 worst anything. That sounds pretty good, but Utah isn't without problems. Figure 5 shows months with the most unhealthy days over the 1999-2023

time period. It showed a staggering 5 unhealthy days in January of 2009. This however has improved as in 2023 it was shown the month with the most unhealthy days was June with 1 day. June also being in the summer where it has been shown that is the season that ranks the worst for air quality (see figure 2)

There are certain counties that have the most unhealthy days in all of Utah. Figure 6 actually shows that this has increased over the years as well. In 1999 the counties that had the highest amount of unhealthy days were Salt Lake and Utah at 2 and 3. In 2009 it was the same two counties at 5 and 5 unhealthy days. In 2019 there was a two way tie for 4 days in Duchesne, Uintah. Staggeringly the highest two counties in 2023 were Duchesne and Uintah again. But at 11 and 18 days respectively. The historical data would point to that something big happened in those counties to cause number changes like that.